

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. (CL) Examination

May 2018

CL 744 Structural Dynamics & Earthquake Engineering

Date: 03.05.2018, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.
5. IS 1893: 2016 Part-I, IS 4326: 1993, IS 13920: 2016, IS 456: 2000 & SP 16 is permitted, provided that they do not contain anything other than the printed matter inside.
6. Assume Fe415 and M20 unless otherwise mentioned.

SECTION – I

Q-1(a) State whether following statements are true or false & explain reasons for the same in brief: [03]

- 1) Stiff structures are better than flexible structures.
- 2) Damages due to earthquake are always allowed in the structures.
- 3) Can my building withstand a magnitude 9.0 earthquake?

Q-1(b) Explain role of shear walls in seismic performance of buildings. [04]

Q-2(a) Figure 1 shows the plan of a single storey building of height 4 m. Assume that the center of mass and center of stiffness of the building coincides with each other having the coordinates as (2.80, 2.5) from point 'a'. The stiffness of each shear wall about its stronger axis is shown in figure in terms of 'k'. Calculate the lateral load shared by each shear walls if the earthquake force of 500 kN acts in the y-direction at center of mass. [04]

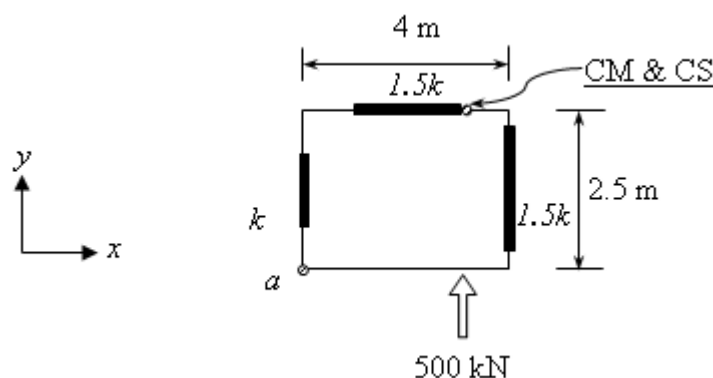


Figure 1

Q-2(b) Figure 2 shows the typical layout and section of a building. Assume slab thickness 110 mm and floor finish as 1.0 kN/m² throughout. All beams have size 250 x 450 mm (including slab), 230 mm thick full height brick masonry wall around outer periphery of building in first floor, no walls in ground floor and 230 mm thick 0.9

m high brick masonry wall around outer periphery of building at roof level. All circular columns of 450 mm diameter and all rectangular columns of 300 x 440 mm size. The building has SMRF and it is hospital building at Changa rested on medium soil. Design the shear reinforcement in beam B₀₅ if DL udl on beam is 40 kN/m and LL udl on beam is 10 kN/m. Use 8 mm dia. 2 legged stirrups.

- Main steel provided in various beams is as follows:

Beam No.	Continuous Top Bars	Extra Top Bars on Left	Extra Top Bars on Right	Continuous Bottom Bars
B ₀₁	2-16 dia.	2-20 dia.	3-20 dia.	3-20 dia.
B ₀₂	2-16 dia.	3-20 dia.	2-20 dia.	3-20 dia.
B ₀₃	2-16 dia.	2-20 dia.	2-20 dia.	4-20 dia.
B ₀₄	2-16 dia.	2-20 dia.	2-20 dia.	3-20 dia.
B ₀₅	2-12 dia.	2-20 dia.	2-20 dia.	4-16 dia.

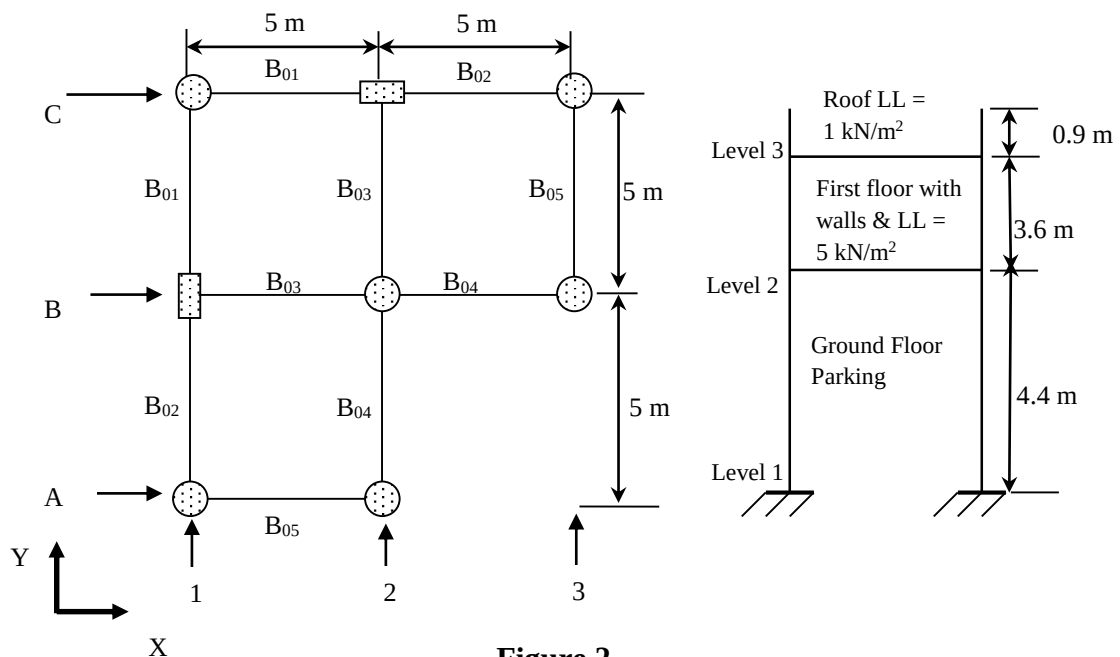


Figure 2

OR

- Q-2(a)** A three storied building has lumped floor weights from bottom to top as $M_1 = M_2 = 1500 \text{ kg}$ and $M_3 = 1000 \text{ kg}$ with storey stiffness of 250 kN/m for all stories. From the free vibration analysis, the natural frequencies and corresponding mode shape coefficients are obtained as follows: **[07]**

$$\omega_1 = 6.33 \text{ rad/sec}, \omega_2 = 17.34 \text{ rad/sec}, \omega_3 = 24 \text{ rad/sec}$$

$$\{\phi_1\} = \{\phi_{11}, \phi_{21}, \phi_{31}\} = \{0.477, 0.84, 1\}$$

$$\{\phi_2\} = \{\phi_{12}, \phi_{22}, \phi_{32}\} = \{1.04, -0.201, 1\}$$

$$\{\phi_3\} = \{\phi_{13}, \phi_{23}, \phi_{33}\} = \{0.893, -1.3, 1\}$$

Consider the building as ordinary office building with ordinary RC moment-resisting frame (OMRF) proposed on medium soil with zone factor $(Z) = \text{III}$.

Calculate the design lateral force at each floor in each mode. Also calculate the storey shear force considering participation of all modes using modal combination method (SRSS or CQC).

- Q-2(b)** **Figure 3** shows the plan of a single storey brick masonry ordinary residential building located in Bhuj. Consider all walls of 230 mm thickness. Decide the position of door (D) & window (W) and maximum size of window that can be provided in wall 'A', 'B' and 'C' considering width of door as 900 mm. Also show the necessary vertical reinforcement to be provided at the critical junctions as a part of strengthening measures. **[07]**

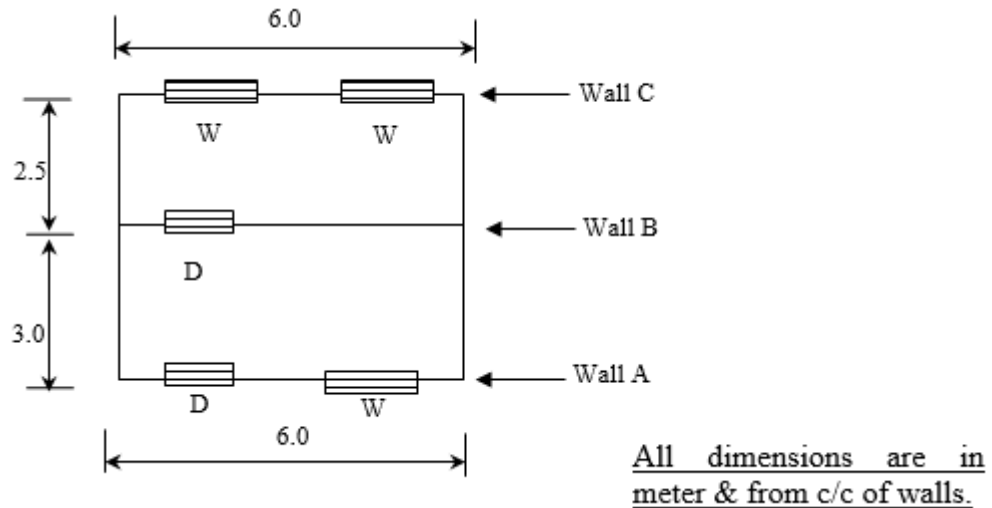


Figure 3

- Q-3(a)** **Figure 4** shows the plan of a single storey building of height 3.5 m. Consider column, $C_A = 300 \text{ mm} \times 500 \text{ mm}$ and $C_B = 500 \text{ mm} \times 300 \text{ mm}$. Consider load intensity of 12 kN/m^2 on area "A" and 23 kN/m^2 on area "B" as shown in figure. Ignore the weight of columns and beams and calculate the centre of mass. Also calculate the size of circular column " C_C " such that the eccentricity in x-direction becomes zero. (Take stiffness of column as $12EI/L^3$ and $E = 22360 \times 10^3 \text{ kN/m}^2$). **[05]**

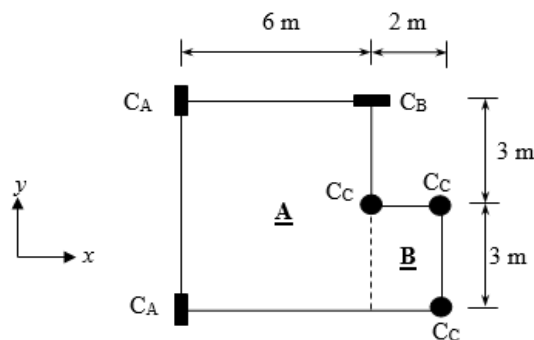
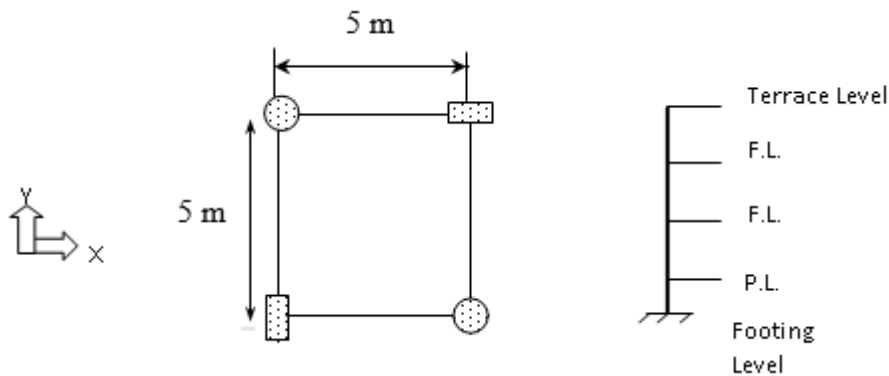


Figure 4

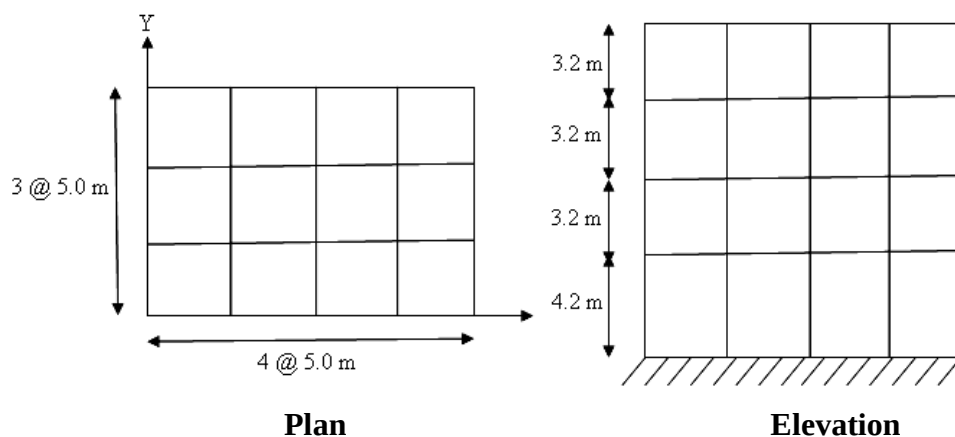
- Q-3(b)** **Figure 5** shows the typical layout and elevation of a residential building proposed to be constructed at V.V.Nagar. Consider the following data: **[09]**

- Slab thickness = 120 mm at all floors

- Storey height = 3 m
 - Live load at floor level = 4 kN/m^2 , roof level = 2 kN/m^2
 - Floor finish = 1 kN/m^2 at all floors
 - Column sizes, $C_A = 600 \text{ mm} \times 300 \text{ mm}$, $C_B = 300 \text{ mm} \times 600 \text{ mm}$, $C_C = 380 \text{ mm}$ dia.
 - Beam sizes, all tie beams (plinth level) = $230 \text{ mm} \times 230 \text{ mm}$ and all other beams = $230 \text{ mm} \times 600 \text{ mm}$.
 - Consider 230 mm thick brick masonry wall of full height on outer periphery of building.
 - Density of RCC = 25 kN/m^3 and Density of brick masonry = 20 kN/m^3
- Calculate the center of stiffness of any intermediate floor level along X & Y direction.

**Figure 5****OR**

- Q-3(b)** Consider a four-storey reinforced concrete office building shown in **Figure 6**. The building is located in Shillong (Seismic zone V). The soil conditions are medium stiff and the entire building is supported on a raft foundation. The R. C. frames are infilled with brick-masonry. The lumped weight due to dead loads is 12 kN/m^2 on floors and 10 kN/m^2 on the roof. The floors are to cater for a live load of 4 kN/m^2 on floors and 1.5 kN/m^2 on the roof. Determine the lateral forces (X & Y directions) at different storey levels by seismic coefficient method. [09]

**Figure 6**

SECTION – II

- Q-4** A uniform cantilever tower of length L has mass per unit length m and flexural rigidity EI . Assuming the shape function $\psi(x) = \frac{3}{2}\left(\frac{x}{L}\right)^2 - \frac{1}{2}\left(\frac{x}{L}\right)^3$, determine its natural frequency [07]
- Q-5(a)** A platform of weight 25000 N is supported on four equal columns which are fixed to the foundation as well as to the platform. Experimentally it is found that a force of 6000 N applied horizontally to the platform produces a displacement of 0.2 cm. It is estimated that damping in the structures is of the order of 5% of the critical damping. Determine (i) undamped natural frequency (ii) absolute damping coefficient (iii) logarithmic decrement (iv) the numbers of cycles and the time required for the amplitude of motion to be reduced from an initial value of 0.2 cm to 0.02 cm. [07]
- Q-5(b)** A shaft of 90 mm diameter and 800 mm long has one of its end is fixed and the other end carries a rotor of weight 6 kN and of radius 200 mm. Determine the natural frequency of transverse and longitudinal vibrations. Take $E = 200 \text{ kN/mm}^2$. [07]

OR

- Q-5(a)** Each pendulum is hanging on an ideal spring with equal mass. The period of vibration for the pendulum is 1.73 sec and 3 sec, respectively. What is the stiffness of the second pendulum with respect to first? [03]
- Q-5(b)** Define: (i) Resonance (ii) Free Vibration (iii) Logarithmic Decrement (iv) Damping ratio [04]
- Q-5(c)** A vertical cantilever of mild steel tube section is 3.0 m long and supports 5 kN weight at its top. The tube has 250 mm external diameter and 5 mm wall thickness. The system is subjected to a harmonic force of 2 kN amplitude and 4 Hz frequency. Find the maximum steady state displacement and bending stress in the tube. Take damping as 5% of critical damping. Take $E = 210 \text{ GPa}$. [07]
- Q-6(a)** Define: Fundamental mode, Natural mode shape [02]
- Q-6(b)** Attempt **Any ONE** of the followings: [12]
- 1)** A two storied building has lumped floor weights from bottom to top as 1000 kg and 500 kg with storey stiffness from bottom to top as 2000 N/m and 1000 N/m. From the free vibration analysis, the natural frequencies and corresponding mode shape coefficients are obtained as follows:
 $\omega_1 = 1 \text{ rad/sec}$, $\omega_2 = 2 \text{ rad/sec}$
 $\{\phi_1\} = \{\phi_{11}, \phi_{21}\} = \{0.5, 1\}$
 $\{\phi_2\} = \{\phi_{12}, \phi_{22}\} = \{-1, 1\}$
 If the 1000 kg and 500 kg masses are displaced to right 1 cm and 3 cm from their steady position respectively and left to vibrate, derive the displacement function of all the masses.

- 2) Calculate the natural frequencies and mode shapes of the two-storey shear building shown in **Figure 7**. Show that the modes of vibration of the above shear building satisfy the orthogonality properties.

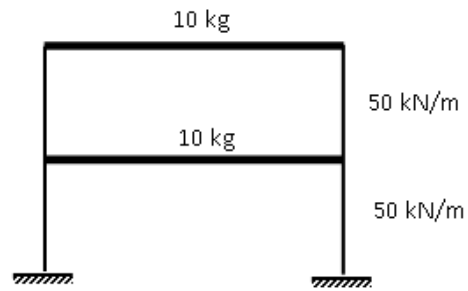


Figure 7
